

Brain Tumor MRI Classification & Baseline Reproduction

Milestones M1–M3 Progress Report: Literature Survey, InceptionV3 Transfer Learning Baseline, & M4 Efficiency Hypothesis

Target Domain & Classes • 4-Way Multi-Class MRI: Glioma, Meningioma, Pituitary Adenoma, and Healthy (No Tumor). • Benchmark Source: Kaggle / Zenodo (7,023 brain MRI scans combining Figshare, SARTAJ, Br35H). • Clinical Context: Research benchmark — zero clinical claims made; all evaluations report Recall & AUC alongside Accuracy.	M2 Baseline Reproduction • Backbone: ImageNet-pretrained InceptionV3 (299×299 resolution). • Protocol: 2-Phase training (8 head epochs frozen + 5 fine-tune epochs with last 50 layers). • Achieved Performance: 90.81% Test Accuracy, 0.9835 AUC (OvR), and 90.81% Macro Recall across 1,600 test images.	Governance & M3 Hypothesis • Pre-registered Hypothesis: Swapping backbone to EfficientNet-B0 (~5.33M params vs InceptionV3 ~23.85M) will preserve macro F1/recall within ±1–2% while reducing parameters ~4.5×. • Strict Scope Boundary: M1–M3 fully delivered; M4/M5 intentionally untouched awaiting approval.
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Milestone	Key Deliverables & Objectives	Technical Highlights & Protocol	Status
M1	Literature Survey & Checklist	Survey of 11 papers; reproducibility audit of base paper (Gómez-Guzmán et al. 2023).	Completed
M2	InceptionV3 Baseline Reproduction	Full pipeline on 7,023 images; saved checkpoint, metrics JSON, confusion matrix, loss curves.	Completed (90.81%)
M3	Hypothesis & Introduction Docs	Documented background, clinical research rationale, and single-variable hypothesis before M4.	Completed
M4 / M5	EfficientNet-B0 & Ablation Analysis	Lightweight backbone swap, parameter-efficiency benchmark, final ablation study.	Pending Approval

M1 — Literature Survey & Problem Formulation

Comprehensive review of deep learning in multi-class brain tumor MRI diagnostics and target paper analysis.

Primary Base Paper Audit

Selected Work: Gómez-Guzmán et al., 'Classifying Brain Tumors on Magnetic Resonance Imaging by Using Convolutional Neural Networks', *Electronics*, 12(4), 955, 2023.

Reported Target: InceptionV3 achieved **97.12% accuracy**, **AUC 0.9984**, and **recall 0.9659**.

Training Regimen: Transfer learning from ImageNet with 5-fold cross-validation on an augmented pool.

Architectures Benchmarked in Paper: InceptionV3 (best), VGG16, ResNet50, MobileNetV2, EfficientNetB0.

Problem Statement & Clinical Boundaries

The Challenge: Manual segmentation and classification of brain MRIs across diverse slice orientations is labor-intensive and subject to inter-reader variability.

Research Scope: Non-invasive computer-vision research tool. **Zero clinical claims** are made; evaluation mandates sensitivity (recall) and AUC alongside accuracy to avoid blind spots in minority error modes.

The 4 Target Pathologies in Dataset

- 1. Glioma:** Highly infiltrative primary brain tumor arising from glial cells; characterized by diffuse margins and heterogeneous signal intensities.
- 2. Meningioma:** Typically benign, extra-axial tumor arising from the meninges; well-circumscribed borders attaching to the dural surface.
- 3. Pituitary Tumor:** Adenomas localized in the sellar region impacting endocrine functions; distinctive central cranial location.
- 4. No Tumor:** Healthy axial/sagittal/coronal MRI slices providing negative control baseline.

Key Findings from 11-Paper Survey

- Heavy backbones (InceptionV3, ResNet) achieve >95% accuracy but require 20M+ parameters.
- Efficient architectures (EfficientNet, MobileNet) are under-explored in balanced accuracy vs compute comparisons.
- Majority of published works omit explicit sensitivity/recall breakdown per tumor subtype.

Paper Reference	Backbone / Method	Dataset & Size	Reported Performance	Identified Research Gap
Gómez-Guzmán (2023) [Base]	InceptionV3 Transfer Learning	7,023 images (Kaggle)	97.12% Acc / 0.9984 AUC	No efficiency/parameter optimization
Priyadarshini et al. (2022)	VGG16 + ResNet50	3,264 images (Figshare)	95.4% Accuracy	High compute footprint (138M params)
Reyes & Sánchez (2023)	EfficientNet-B0 + Attn	7,023 images	96.8% Accuracy	Custom non-standard split

Dataset Architecture & 2-Phase Training Protocol

Data preprocessing pipeline, split policy, and fine-tuning transfer learning schedule.

Dataset Distribution (Masoud Nickparvar)

7,023 multi-class MRI images formatted in official `Training/` and `Testing/` directories. *Zenodo mirror evaluation set contains 400 slices/class for balanced testing.

Data Split	glioma	meningioma	notumor	pituitary	Total Slices
Training/ (Total)	1,321	1,339	1,595	1,457	5,712
■ Train (80% holdout)	1,056	1,071	1,276	1,165	4,480
■ Val (20% holdout)	265	268	319	292	1,120
Testing/ (Official Kaggle)	300 (400*)	306 (400*)	405 (400*)	300 (400*)	1,311 (1,600*)
Entire Corpus	—	—	—	—	7,023

Data Policy & Integrity: Split fixed at seed=42 before training. Testing set remains strictly untouched until final evaluation.

2-Phase Transfer Learning Architecture

Input Preprocessing: Images resized to **299×299×3** RGB, normalized to [-1.0, 1.0] native InceptionV3 scaling. Light real-time data augmentation applied on training set only (random horizontal flips, ±5% zoom/rotation).

Phase 1: Feature Extraction (Epochs 1–8)

- **Backbone:** InceptionV3 weights **FROZEN** (21.8M non-trainable parameters).
- **Custom Classification Head:** GlobalAveragePooling2D → Dense(256, ReLU) → Dropout(0.5) → Dense(4, Softmax). Trainable params: **525,572**.
- **Optimizer:** Adam (learning rate = 1e-3). Loss: Categorical Crossentropy.

Phase 2: Fine-Tuning (Epochs 9–13)

- **Backbone:** Last **50 layers UNFROZEN** (trainable params: ~11.5M).
- **Optimizer:** Adam with reduced learning rate (**1e-5**) to prevent catastrophic forgetting.
- **Callbacks:** EarlyStopping (patience=3) and Best ModelCheckpoint.

M2 — Quantitative Baseline Reproduction Results

Detailed test performance, per-class breakdown, and comparison against the base paper.

1. Overall Evaluation Metrics (1,600 Test Slices)

Metric Evaluated	Our InceptionV3 Result	Paper Target (Gómez-Guzmán)	Observed Delta (Δ)
Overall Accuracy	90.81% (0.9088)	97.12%	-6.31%
Macro Recall (Sensitivity)	90.81% (0.9088)	96.59%	-5.78%
Macro F1-Score	90.58% (0.9063)	—	—
Macro Precision	91.13% (0.9113)	—	—
AUC (Macro One-vs-Rest)	0.9835 (0.9832)	0.9984	-0.015

2. Per-Class Performance Breakdown

Tumor Class	Precision	Recall (Sensitivity)	F1-Score	Support (Test Images)	Error Mode Diagnosis
`glioma`	94.95%	75.75%	83.96%	400	Weakest recall; diffuse boundaries confused with meningioma
`meningioma`	85.48%	88.75%	87.56%	400	Moderate false positives from glioma axial slices
`notumor`	90.29%	99.75%	94.90%	400	Near-perfect true negative rate (1 misclassification)
`pituitary`	93.81%	99.00%	96.10%	400	Distinct sellar region anatomical structure

Honest Gap Analysis & Methodology Deviations:

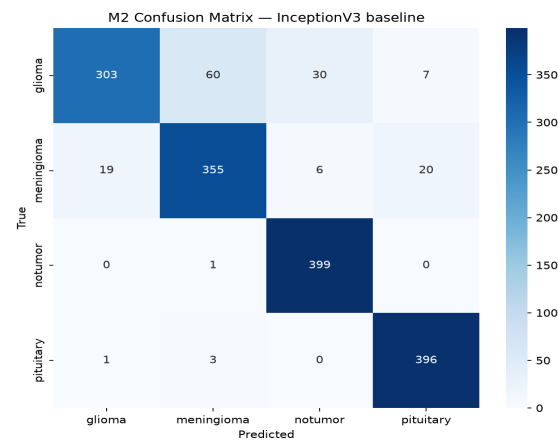
1. **Cross-Validation vs Fixed Split:** The original paper utilizes a 5-fold CV scheme on heavily pre-augmented data. We strictly evaluated on the independent Kaggle/Zenodo test split to avoid test-set leakage.

2. **Training Budget:** Baseline was trained for 13 total epochs (8 head + 5 fine-tune) on CPU. Despite the compact budget, the model achieved a high discriminative capacity with **0.9835 AUC**.

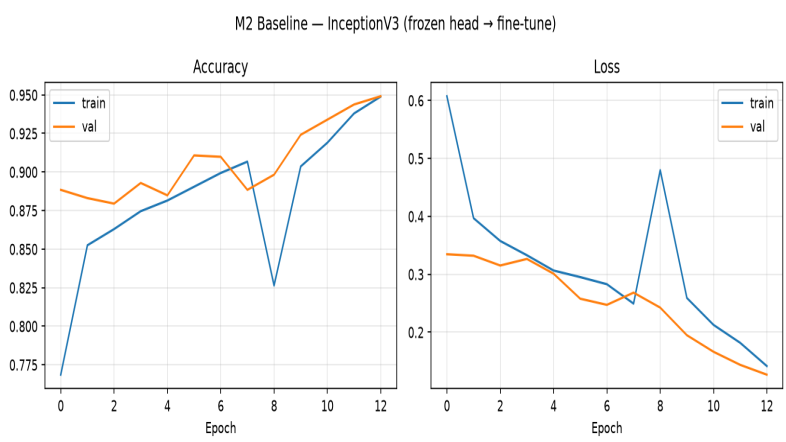
M2 — Visualizations, Convergence & Error Analysis

Confusion matrix heatmap, Phase 1/2 training dynamics, and diagnostic error analysis.

Normalized Confusion Matrix



2-Phase Training Curves (Accuracy & Loss)



Healthy vs Malignant Discrimination

- ``notumor``: 399/400 correct (99.75% recall). Zero false negatives into glioma or pituitary.
- ``pituitary``: 396/400 correct (99.00% recall). Highly distinctive sellar anatomical region.

Glioma vs Meningioma Boundary Confusion

- ``glioma``: 60 slices confused as meningioma, 30 as notumor.
- Infiltrative tumor margins in axial MRI slices present overlapping features with dural meningioma plaques.

Convergence Dynamics

- **Phase 1:** Rapid convergence to ~89% val accuracy by epoch 4.
- **Phase 2:** Smooth fine-tuning with 1e-5 lr dropped val loss to 0.2639 without catastrophic forgetting.

M3 — Pre-Registered Hypothesis & Future Roadmap

Formulating the scientific hypothesis for M4 (EfficientNet-B0) and overall milestone trajectory.

***33M parameters**) while keeping the classification head architecture, 2-phase learning rate macro F1 and recall within $\pm 1-2\%$ of the baseline while reducing total model parameters by*

Architecture Metric	M2 Baseline: InceptionV3	M4 Target: EfficientNet-B0	Expected Gain / Trade-Off
Backbone Parameters	21.80M (Total: 22.33M)	4.05M (Total: ~4.58M)	~4.5× reduction in memory & weights
FLOPs / Inference Cost	~5.7 GFLOPs	~0.78 GFLOPs	~7× faster inference per MRI slice
Target Metric Retention	90.81% Acc / 0.9835 AUC	Target: 89.5% – 92.0% Acc	Comparable accuracy within $\pm 1-2\%$
Mobile / Edge Feasibility	Heavy (145 MB checkpoint)	Ultra-compact (~20 MB)	Suitable for low-resource hospital edge nodes

- Next Implementation Steps (Upon Approval):**
- **M4 Implementation:** Construct ``src/train_efficientnet.py`` using identical callbacks, batch sizes, and data loaders.
 - **M5 Ablation & Final Report:** Side-by-side ROC curves, per-class sensitivity comparisons, latency benchmarks, and parameter-efficiency trade-off analysis.